Network Layer Functions

Prof. Neminath Hubballi

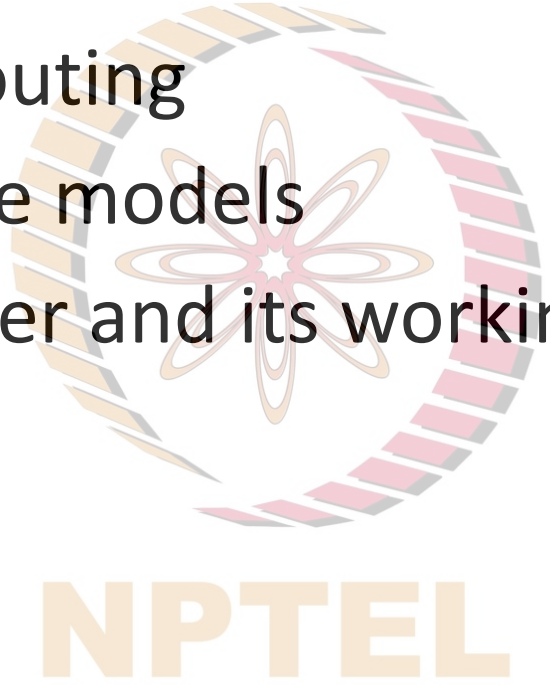
Department of Computer Science and Engineering

Indian Institute of Technology Indore

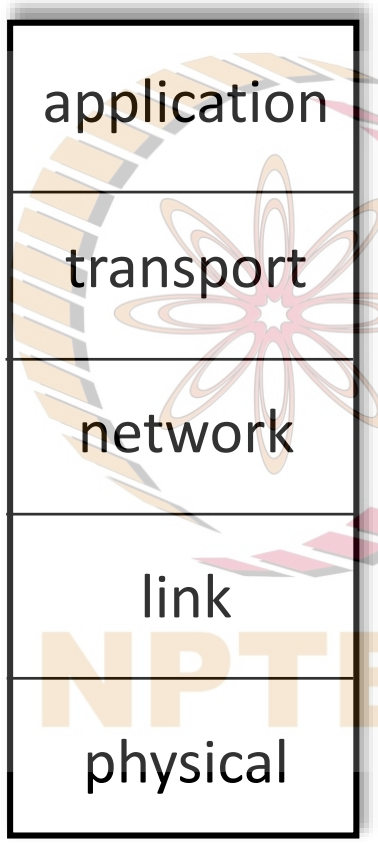
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Overview

- Forwarding versus routing
- Network layer service models
- What is inside a router and its working



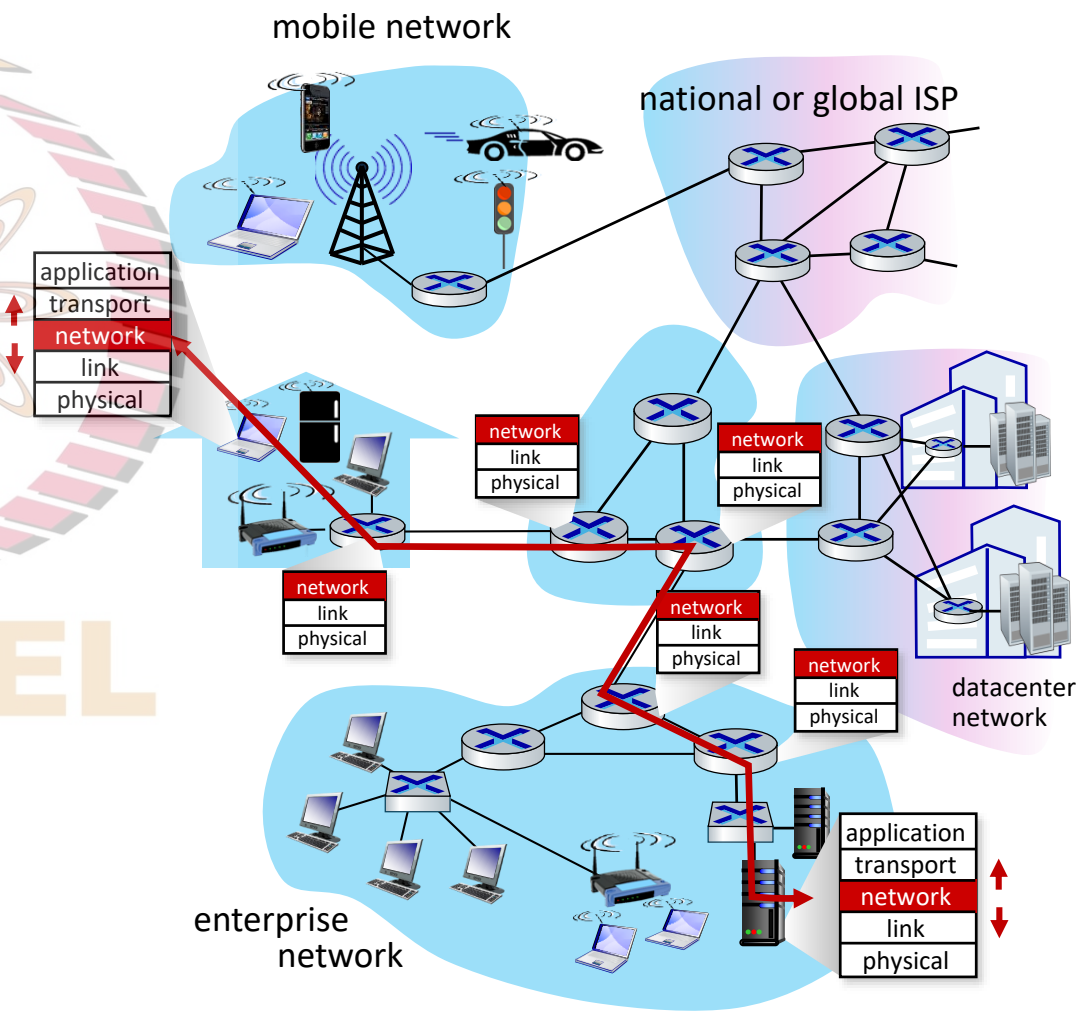
Services, Layering and Encapsulation



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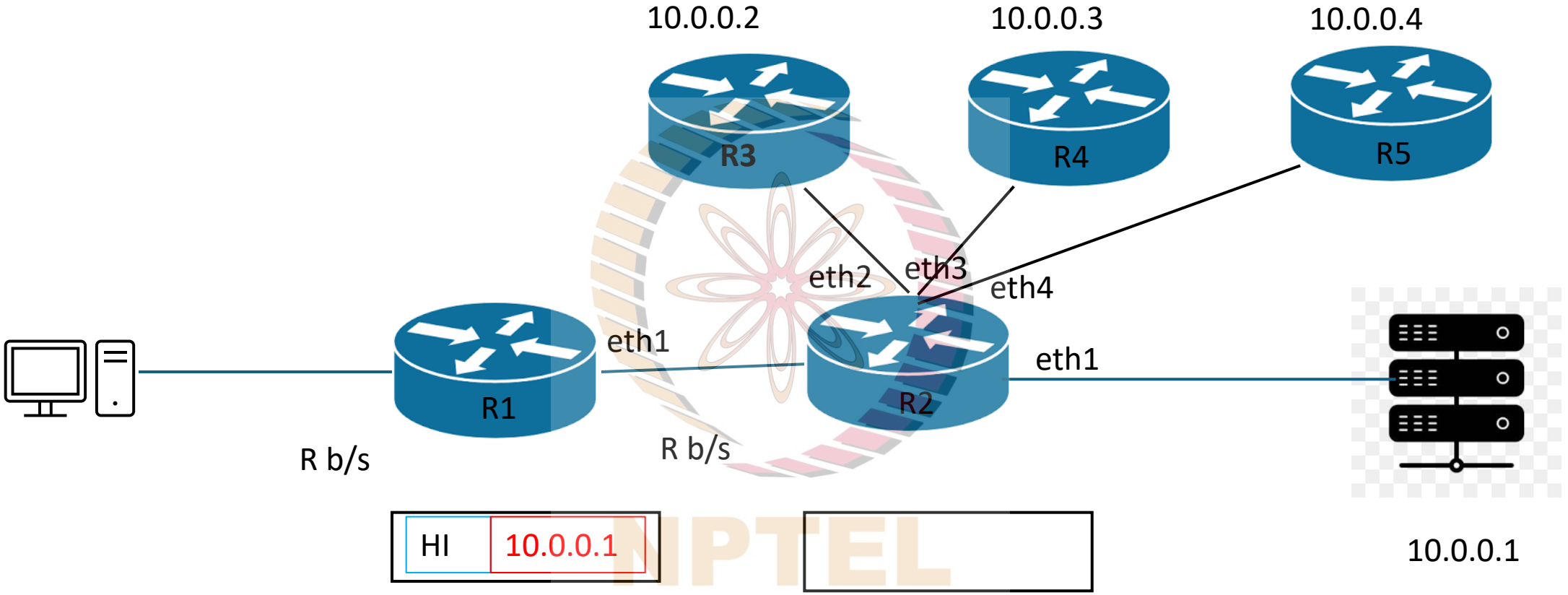
Network-layer Services and Protocols

- Transport segment from sending to receiving host
  - sender: encapsulates segments into datagrams, passes to link layer
  - receiver: delivers segments to transport layer protocol
- Network layer protocols in *every Internet device*: hosts, routers
- router:
  - examines header fields in all IP datagrams passing through it
  - moves datagrams from input ports to output ports to transfer datagrams along end-end path



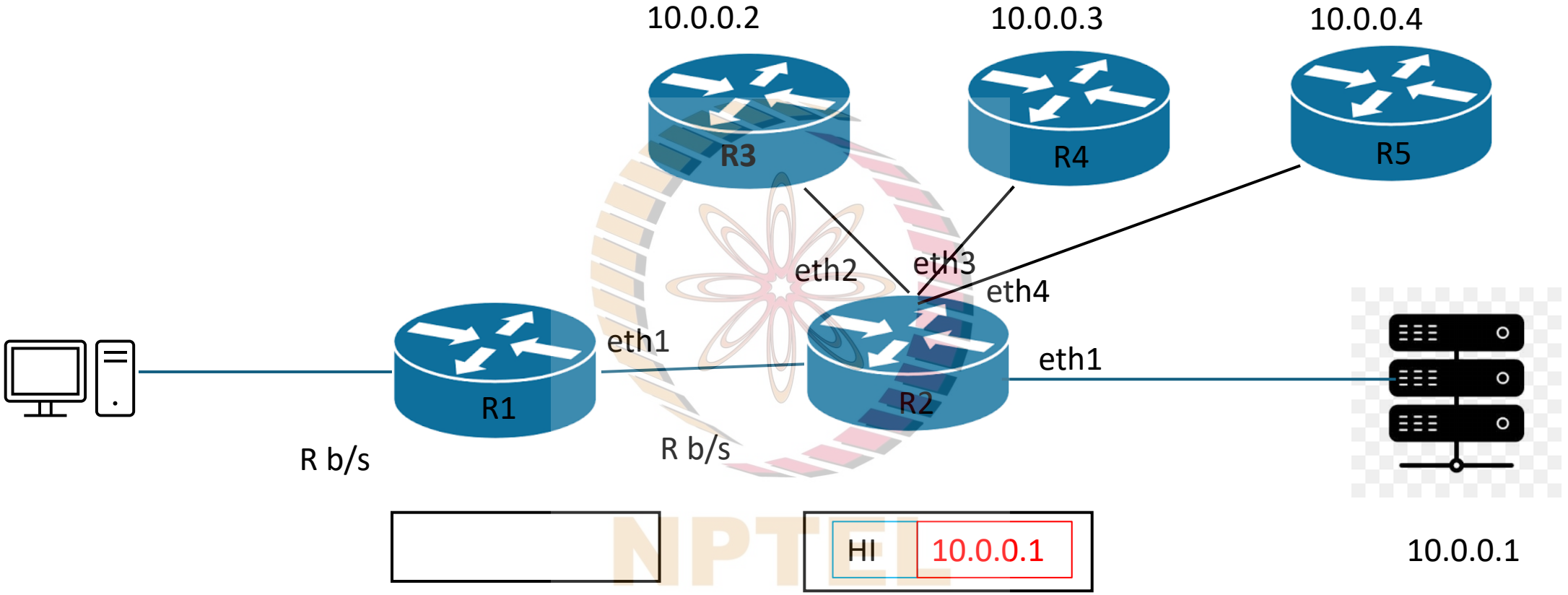
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Forwarding Operation



Destination IP	Next hop	Interface
10.0.0.2	R3	eth2
10.0.0.3	R4	eth3
10.0.0.4	R5	eth4
10.0.0.1	--	eth1

Forwarding Operation



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Forwarding v/s Routing

Network-layer functions:

- *forwarding:* move packets from a router's input link to appropriate router output link
- *routing:* determine route taken by packets from source to destination
  - *routing algorithms*

Analogy: taking a trip

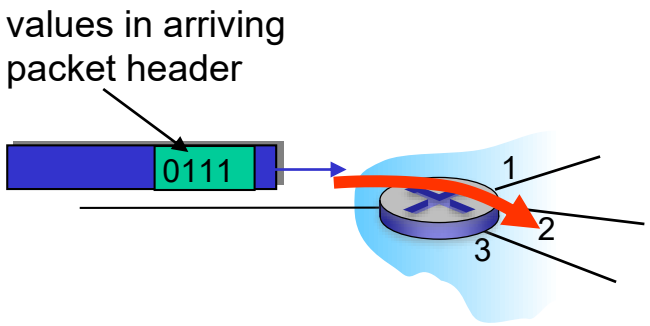
- *forwarding:* process of getting through single interchange
- *routing:* process of planning trip from source to destination

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Network Layer: Data Plane and Control Plane

Data plane:

- *local*, per-router function
- determines how datagram arriving on router input port is forwarded to router output port



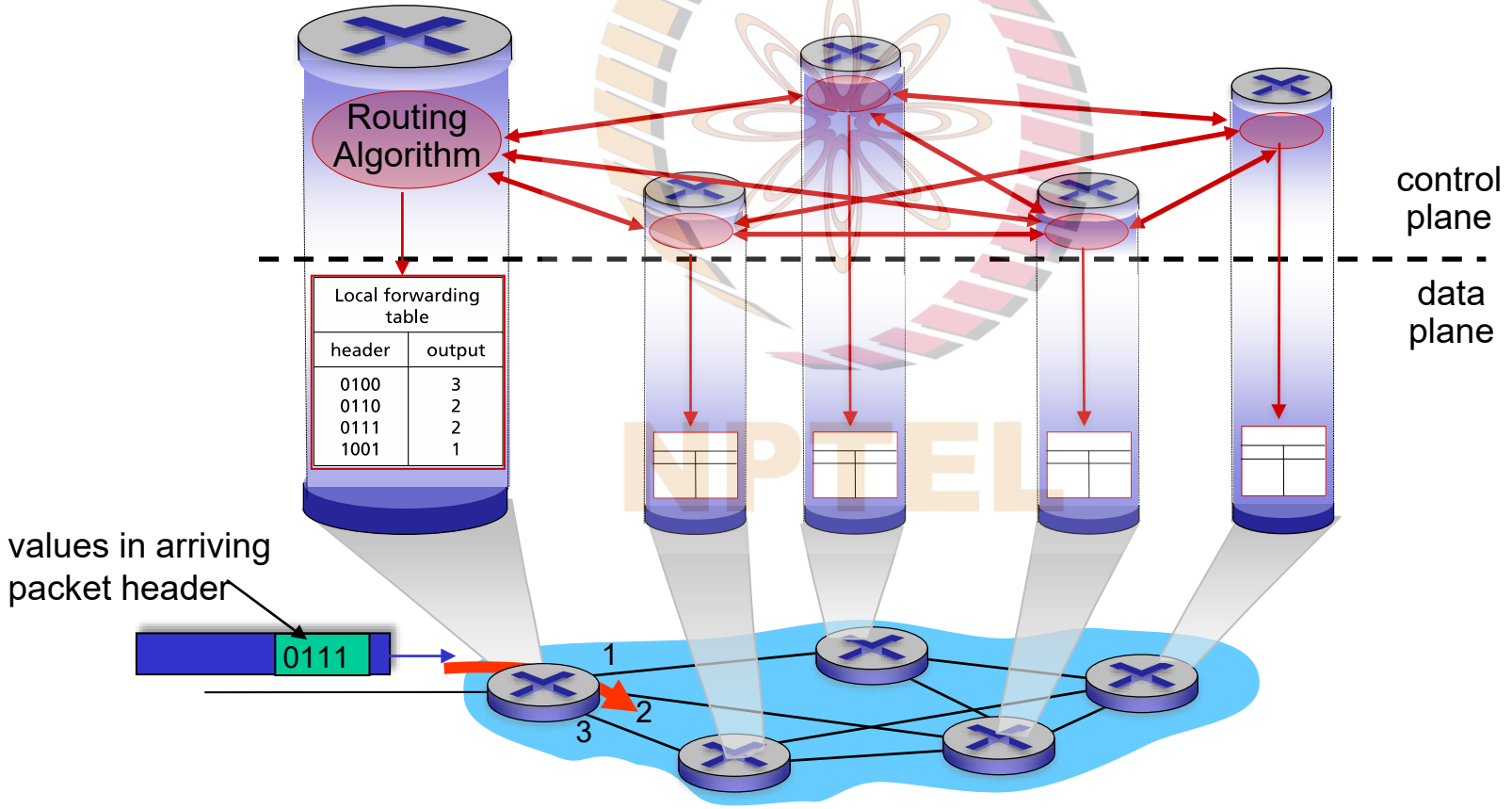
Control plane

- *network-wide* logic
- determines how datagram is routed among routers along end-end path from source host to destination host

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Per-Router Control Plane

Individual routing algorithm components *in each and every router* interact in the control plane



Network Service Models

- On individual datagrams
  - Guaranteed delivery
  - Time Bounded guaranteed delivery
- On flows
  - In order delivery
  - Guaranteed minimal bandwidth
  - Restrictions on interpacket services

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Network-Layer Service Model

Network Architecture	Service Model	Quality of Service (QoS) Guarantees ?			
		Bandwidth	Loss	Order	Timing
Internet	best effort	none	no	no	no

Internet "best effort" service model

No guarantees on:

i. successful datagram delivery to destination
ii. timing or order of delivery
iii. bandwidth available to end-end flow

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Network-Layer Service Models

Network Architecture	Service Model	Quality of Service (QoS) Guarantees ?			
		Bandwidth	Loss	Order	Timing
Internet	best effort	none	no	no	no
ATM	Constant Bit Rate	Constant rate	yes	yes	yes
ATM	Available Bit Rate	Guaranteed min	no	yes	no
Internet	Intserv Guaranteed (RFC 1633)	yes	yes	yes	yes
Internet	Diffserv (RFC 2475)	possible	possibly	possibly	no

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Reflections on Best-Effort Service

- Simplicity of mechanism has allowed Internet to be widely deployed adopted
- Sufficient provisioning of bandwidth allows performance of real-time applications (e.g., interactive voice, video) to be "good enough" for "most of the time"
- Replicated, application-layer distributed services (datacenters, content distribution networks) connecting close to clients' networks, allow services to be provided from multiple locations
- Congestion control of "elastic" services helps

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